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Micro light-emitting diode device

Abstract

A micro light-emitting diode device includes a substrate, a micro light-emitting diode, and a transparent top electrode. The micro light-emitting diode is disposed on the substrate and includes a p-type GaN layer, an n-type III-nitride layer above the p-type GaN layer, an n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer above and in contact with the n-type III-nitride layer, and an active layer between the p-type GaN layer and the n-type III-nitride layer. x is equal to or greater than about 0.02. The transparent top electrode covers and is in contact with the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer. A refractive index of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer is smaller than a refractive index of the n-type III-nitride layer. A sum of the thicknesses of the n-type III-nitride layer and the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer is greater than a sum of the thicknesses of the active layer and the p-type GaN layer.

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Background/Summary

BACKGROUND

Field of Invention

(1) The present disclosure relates to a micro light-emitting diode device.

Description of Related Art

(2) The statements in this section merely provide background information related to the present disclosure and do not necessarily constitute prior art.

(3) As a light source, light-emitting diodes (LEDs) have many advantages, including low energy consumption, long lifetime, small size, and fast switching. Hence, conventional lighting, such as incandescent lighting, is gradually replaced by LED lights. The properties regarding LEDs also fit applications on displays. Researches on displays using micro light-emitting devices, or specifically, micro light-emitting diodes (micro LEDs), have become popular in recent years. Commercial lighting applications made of micro LEDs are nearly within reach.

(4) GaN (InGaN, AlGaN) based LEDs have become the majority of LED Illumination. Since the GaN based LED display gradually becomes mainstream in the display market, searching for better performance for the GaN based LED display becomes an important topic, and the display industries are starting to pay attention on it.

SUMMARY

(5) According to some embodiments of the present disclosure, a micro light-emitting diode device includes a substrate, a micro light-emitting diode, and a transparent top electrode. The micro light-emitting diode has a lateral width smaller than 100 μm. The micro light-emitting diode is disposed on the substrate and includes a p-type GaN layer, an n-type III-nitride layer, an n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer, and an active layer. The n-type III-nitride layer is above the p-type GaN layer. III refers to at least one element of aluminum, gallium, and indium. The n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer is above and in contact with the n-type III-nitride layer, in which x is equal to or greater than about 0.02. A thickness of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer is greater than 10 nm. The active layer is between the p-type GaN layer and the n-

type III-nitride layer. The transparent top electrode at least partially covers and is in contact with the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer. A refractive index of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer is smaller than a refractive index of the n-type III-nitride layer. A sum of thicknesses of the p-type GaN layer, the active layer, the n-type III-nitride layer, and the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer is greater than 400 nm. A sum of the thicknesses of the n-type III-nitride layer and the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer is greater than a sum of the thicknesses of the active layer and the p-type GaN layer.

(6) It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:
- (2) FIG. 1 is a schematic cross-sectional view of a micro light-emitting diode device according to some embodiments of the present disclosure;
- (3) FIG. 2 is a schematic diagram showing two structures;
- (4) FIG. 3 is a partial schematic cross-sectional view of a transparent top electrode according to some embodiments of the present disclosure;
- (5) FIG. 4 is a schematic cross-sectional view of a micro light-emitting diode device according to some embodiments of the present disclosure;
- (6) FIG. 5 is a schematic cross-sectional view of a micro light-emitting diode device according to some embodiments of the present disclosure;
- (7) FIG. 6 is a schematic cross-sectional view of a micro light-emitting diode device according to some embodiments of the present disclosure;
- (8) FIG. 7 is a schematic cross-sectional view of a micro light-emitting diode device according to some embodiments of the present disclosure;
- (9) FIG. 8 is a schematic cross-sectional view of a micro light-emitting diode device according to some embodiments of the present disclosure;
- (10) FIG. 9 is a schematic top view of a micro light-emitting diode device according to some embodiments of the present disclosure; and
- (11) FIG. 10 is a schematic top view of a micro light-emitting diode device according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

- (12) Reference will now be made in detail to the present embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.
- (13) In various embodiments, description is made with reference to figures. However, certain embodiments may be practiced without one or more of these specific details, or in combination with other known methods and configurations. In the following description, numerous specific details are set forth, such as specific configurations, dimensions, and processes, etc., in order to provide a thorough understanding of the present disclosure. In other instances, well-known semiconductor processes and manufacturing techniques have not been described in particular detail in order to not unnecessarily obscure the present disclosure. Reference throughout this specification to “one embodiment,” “an embodiment”, “some embodiments” or the like means that a particular feature, structure, configuration, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Thus, the appearances of the phrase “in one

embodiment,” “in an embodiment”, “according to some embodiments” or the like in various places throughout this specification are not necessarily referring to the same embodiment of the disclosure. Furthermore, the particular features, structures, configurations, or characteristics may be combined in any suitable manner in one or more embodiments.

(14) The terms “micro” device, “micro” p-n diode or “micro” LED as used herein may refer to the descriptive size of certain devices or structures according to embodiments of the present disclosure. As used herein, the terms “micro” devices or structures may be meant to refer to the scale of 1 to 100 μm . However, it is to be appreciated that embodiments of the present disclosure are not necessarily so limited, and that certain aspects of the embodiments may be applicable to larger, and possibly smaller size scales. The refractive index mentioned in the following, unless otherwise specified, is based on the refractive index of light with a wavelength λ of 550 nm.

(15) Reference is made to FIG. 1. FIG. 1 is a schematic cross-sectional view of a micro light-emitting diode device **100A** according to some embodiments of the present disclosure. As shown in FIG. 1, in some embodiments, the micro light-emitting diode device **100A** includes a substrate **110**, a micro light-emitting diode **120**, and a transparent top electrode **130**. The micro light-emitting diode **120** has a lateral width smaller than 100 μm . For example, the lateral width is the width of micro light-emitting diode **120** shown in FIG. 1. The substrate **110** has a metal pad **111** thereon. The micro light-emitting diode **120** is disposed on the substrate **110** and includes a p-type GaN layer **121**, an n-type III-nitride layer **123**, an n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124**, and an active layer **122**. The n-type III-nitride layer **123** is above the p-type GaN layer **121**. “III” (group III) of the n-type III-nitride layer **123** refers to (or consists of) at least one element of aluminum, gallium, and indium. The n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** is above and in contact with the n-type III-nitride layer **123**. It should be pointed out that “Al” of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** refers to aluminum. “Ga” of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** refers to gallium. “In” of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** refers to indium. “N” of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** refers to nitrogen. The Al composition x is equal to or greater than about 0.02. In other words, the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** contains a content of aluminum equal to or greater than 2 wt %. The Ga composition y is a positive number. The active layer **122** is between the p-type GaN layer **121** and the n-type III-nitride layer **123**. The transparent top electrode **130** at least partially covers and is in contact with the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124**.

(16) It should be pointed out that since the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** contains aluminum relative to the n-type III-nitride layer **123**, the band gap of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** is larger than the band gap of the n-type III-nitride layer **123**. In general, the refractive index of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** with the larger band gap will be smaller than the refractive index of the n-type III-nitride layer **123** with the smaller band gap. In some embodiments, a refractive index of the transparent top electrode **130** is smaller than the refractive index of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124**. In other words, the refractive index of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** is between the refractive index of the n-type III-nitride layer **123** and the refractive index of the transparent top electrode **130**. According to Snell's Law, the phenomenon of refraction occurs when light propagates from one medium to another medium with a different refractive index. When the difference in refractive index between the two mediums is large, more light will be reflected at the interface between the two mediums. In detail, assuming that light travels from an optically dense medium (e.g., the n-type III-nitride layer **123**) with a refractive index of n_1 to an optically sparser medium (e.g., the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124**) with a refractive index of n_2 smaller than n_1 , and the incident angle θ_1 is equal to the critical angle θ_c , the refracted ray will proceed along the tangent of the interface. The critical angle θ_c is the smallest angle of incidence at which total internal reflection occurs and can be determined by the following equation (1).

(17) $c = \sin^{-1}(n_2 / n_1)$ (1)

(18) According to the above equation, it can be seen that the refractive index of n_2 with a larger value can make the critical angle θ_c larger. That is, less light will be reflected at the interface between the two mediums. Reference is made to FIG. 2. FIG. 2 is a schematic diagram showing two structures, in which the left structure includes the n-type III-nitride layer **123** and the transparent top electrode **130** stacked thereon, and the right structure includes the n-type III-nitride layer **123**, the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124**, and the transparent top electrode **130** sequentially stacked. Since the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** of the right structure in FIG. 2 has a larger refractive index than the transparent top electrode **130** of the left structure in FIG. 2, the right structure has a larger critical angle θ_c than the left structure. Therefore, compared with the left structure with only the n-type III-nitride layer **123** and the transparent top electrode **130** stacked, the right structure with the n-type III-nitride layer **123**, the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124**, and the transparent top electrode **130** can have better light extraction efficiency.

(19) In some embodiments, the refractive index of the n-type III-nitride layer **123** may be 2.5, and the refractive index of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** is between 2.0 and 2.5. Since the difference in refractive index between the n-type III-nitride layer **123** and the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** is appropriately small, more light propagating in the n-type III-nitride layer **123** can enter the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** (i.e., less light is reflected at the interface between the n-type III-nitride layer **123** and the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124**). In this way, the light extraction efficiency of the micro light-emitting diode **120** can be effectively improved.

(20) In some embodiments, the thickness of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** may be greater than 10 nm. In this way, the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** may have sufficient thickness to appear the effect of refractive index matching.

(21) In some embodiments, the substrate **110** having the metal pad **111** thereon is a receiving substrate. For example, the receiving substrate may be, but is not limited to, a display substrate.

(22) In some embodiments, the thickness of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** may be greater than 80 nm. In this way, the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** with the larger thickness can make the effect of refractive index matching appear more significantly. In addition, the flexibility of the processes of manufacturing the micro light-emitting diode **120** can be significantly increased.

(23) In some embodiments, a sum of thicknesses of the p-type GaN layer **121**, the active layer **122**, the n-type III-nitride layer **123**, and the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** is greater than 400 nm. In this way, the micro light-emitting diode **120** can have sufficient strength to cope with subsequent transferring processes.

(24) In some embodiments, a sum of the thicknesses of the n-type III-nitride layer and the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer is greater than a sum of the thicknesses of the active layer and the p-type GaN layer. In this way, the insulation distance between the transparent top electrode **130** and the metal pad **111** can be increased to reduce the possibility of electrical short.

(25) In some embodiments, the thickness of the n-type III-nitride layer is greater than the sum of the thickness of the active layer and the thickness of the p-type GaN layer. In this way, the insulation distance between the transparent top electrode **130** and the metal pad **111** can also be increased to reduce the possibility of electrical short.

(26) In some embodiments, the thickness of the n-type III-nitride layer is greater than 0.5 μm . It makes the isolation layer **140** easier to be thicker to avoid the electrical short.

(27) In some embodiments, the transparent top electrode **130** is made of transparent conductive oxide. For example, the transparent conductive oxide may include ITO (Indium Tin Oxide), IZO (indium tin oxide), and AZO (aluminum zinc oxide), but the disclosure is not limited in this regard.

In this way, more light propagating in the n-type III-nitride layer **123** can enter the transparent top electrode **130** via the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** with less reflection, such that the light extraction efficiency of the micro light-emitting diode **120** can be further improved.

(28) Reference is made to FIG. **3**. FIG. **3** is a partial schematic cross-sectional view of a transparent top electrode **130** according to some embodiments of the present disclosure. As shown in FIG. **3**, in some embodiments, the transparent top electrode **130** includes nano metal wires **131** and an encapsulation material **132** in which the nano metal wires **131** are embedded. The refractive index of the encapsulation material **132** can be regarded as the refractive index of the transparent top electrode **130**. The refractive index of the encapsulation material **132** is smaller than the refractive index of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124**. In this way, more light propagating in the n-type III-nitride layer **123** can enter the encapsulation material **132** of the transparent top electrode **130** via the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** with less reflection, such that the light extraction efficiency of the micro light-emitting diode **120** can be further improved.

(29) As shown in FIG. **1**, in some embodiments, the micro light-emitting diode device **100A** further includes an isolation layer **140**. The isolation layer **140** is on the substrate **110** and surrounds the micro light-emitting diode **120**. A height H2 of the isolation layer **140** relative to the substrate **110** is equal to or smaller than a height H1 of a top surface of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** relative to the substrate **110**. In this way, the contact area between the micro light-emitting diode **120** and the transparent top electrode **130** is larger, so the resistance can be smaller. In addition, the upward light-emitting efficiency of the micro light-emitting diode device **100A** can be better (because the isolation layer **140** does not cover the top surface of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124**).

(30) In some embodiments, a refractive index of the isolation layer **140** is smaller than the refractive index of the transparent top electrode **130**. That is, the difference in refractive index between the n-type III-nitride layer **123** and the isolation layer **140** will be greater than the difference in refractive index between the n-type III-nitride layer **123** and the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124**. In this way, more light will be reflected at the interface between the n-type III-nitride layer **123** and the isolation layer **140** than the interface between the n-type III-nitride layer **123** and the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124**, so that the upward light-emitting efficiency of the micro light-emitting diode device **100A** can be further improved.

(31) In some embodiments, the isolation layer **140** includes resin, but the disclosure is not limited in this regard.

(32) Reference is made to FIG. **4**. FIG. **4** is a schematic cross-sectional view of a micro light-emitting diode device **100B** according to some embodiments of the present disclosure. As shown in FIG. **4**, in some embodiments, the micro light-emitting diode device **100B** includes a substrate **110**, a micro light-emitting diode **120**, a transparent top electrode **130**, and an isolation layer **140**. A difference between the embodiments shown in FIGS. **1** and **4** is that a height H3 of the isolation layer **140** as shown in FIG. **4** relative to the substrate **110** is greater than the height H1 of the top surface of the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** relative to the substrate **110**. In this way, the thickness of the isolation layer **140** shown in FIG. **4** is larger (compared to the isolation layer **140** shown in FIG. **1**), which is beneficial to increase the production yield rate of the micro light-emitting diode device **100B**.

(33) Reference is made to FIG. **5**. FIG. **5** is a schematic cross-sectional view of a micro light-emitting diode device **100C** according to some embodiments of the present disclosure. As shown in FIG. **5**, in some embodiments, the micro light-emitting diode device **100C** includes a substrate **110**, a micro light-emitting diode **120**, a transparent top electrode **130**, and an isolation layer **140**. A difference between the embodiments shown in FIGS. **1** and **5** is that the micro light-emitting diode device **100C** as shown in FIG. **5** further includes an encapsulation layer **150**. The encapsulation

layer **150** is above and in contact with the transparent top electrode **130**. Specifically, the encapsulation layer **150** encapsulates the micro light-emitting diode **120**, the transparent top electrode **130**, and the isolation layer **140**. A refractive index of the encapsulation layer **150** is smaller than the refractive index of the transparent top electrode **130**. In this way, more light propagating in the n-type III-nitride layer **123** can enter the encapsulation layer **150** sequentially via the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** and the transparent top electrode **130** with less reflection, such that the light extraction efficiency of the micro light-emitting diode **120** can be further improved.

(34) Reference is made to FIG. 6. FIG. 6 is a schematic cross-sectional view of a micro light-emitting diode device **100D** according to some embodiments of the present disclosure. As shown in FIG. 6, in some embodiments, the micro light-emitting diode device **100D** includes a substrate **110**, a micro light-emitting diode **120**, a transparent top electrode **130**, and an isolation layer **140**. A difference between the embodiments shown in FIGS. 1 and 6 is that the micro light-emitting diode device **100D** as shown in FIG. 6 further includes a dielectric layer **160**. The dielectric layer **160** is between the isolation layer **140** and a side surface of the micro light-emitting diode **120**. The dielectric layer **16** can reduce surface recombination on the side surface of the micro light-emitting diode **120** and thus it may improve the radiative recombination rate of the micro light-emitting diode **120**.

(35) Reference is made to FIG. 7. FIG. 7 is a schematic cross-sectional view of a micro light-emitting diode device **100E** according to some embodiments of the present disclosure. As shown in FIG. 7, in some embodiments, the micro light-emitting diode device **100E** includes a substrate **110**, a micro light-emitting diode **120**, a transparent top electrode **130**, and an isolation layer **140**. A difference between the embodiments shown in FIGS. 1 and 7 is that the micro light-emitting diode device **100E** further includes an MgO layer **170**. The MgO layer **170** covers and is in contact with the transparent top electrode **130**. A refractive index of the MgO layer **170** is smaller than the refractive index of the transparent top electrode **130**. In this way, more light propagating in the n-type III-nitride layer **123** can enter the MgO layer **170** sequentially via the n-doped Al.sub.xGa.sub.yIn.sub.(1-x-y)N layer **124** and the transparent top electrode **130** with less reflection, such that the light extraction efficiency of the micro light-emitting diode **120** can be further improved.

(36) Reference is made to FIG. 8. FIG. 8 is a schematic cross-sectional view of a micro light-emitting diode device **100F** according to some embodiments of the present disclosure. As shown in FIG. 8, in some embodiments, the micro light-emitting diode device **100F** includes a substrate **110**, a micro light-emitting diode **120**, a transparent top electrode **130**, an isolation layer **140**, and an MgO layer **170**. A difference between the embodiments shown in FIGS. 7 and 8 is that the transparent top electrode **130** and the MgO layer **170** of the micro light-emitting diode device **100F** have larger coverage areas. In this way, the light output distribution of the micro light-emitting diode device **100F** can be more symmetric.

(37) Reference is made to FIG. 9. FIG. 9 is a schematic top view of a micro light-emitting diode device **100G** according to some embodiments of the present disclosure. As shown in FIG. 9, the micro light-emitting diode device **100G** includes a substrate **110**, a plurality of micro light-emitting diodes **120R**, **120G**, **120B**, and an isolation layer **140**. The micro light-emitting diodes **120R**, **120G**, **120B** are disposed on the substrate **110**. Each of the micro light-emitting diodes **120R**, **120G**, **120B** has a structure similar to that of the micro light-emitting diode **120** shown in FIG. 1 and has a lateral width smaller than 100 μm . The isolation layer **140** is disposed on the substrate **110** and surrounds the micro light-emitting diodes **120R**, **120G**, **120B**. The micro light-emitting diodes **120R**, **120G**, **120B** are configured to emit lights of different wavelengths respectively. For example, the micro light-emitting diodes **120R**, **120G**, **120B** are configured to emit a red light, a green light, and a blue light. Since the micro light-emitting diodes **120R**, **120G**, **120B** are surrounded by the same isolation layer **140**, lights emitted by the micro light-emitting diodes **120R**,

120G, 120B can be evenly mixed.

(38) In some embodiments, a distance of any two of the micro light-emitting diodes **120R, 120G, 120B** is less than 2 mm. In this way, lights emitted by the micro light-emitting diodes **120R, 120G, 120B** can be mixed more evenly.

(39) Reference is made to FIG. **10**. FIG. **10** is a schematic top view of a micro light-emitting diode device **100H** according to some embodiments of the present disclosure. Compared to the micro light-emitting diode device **100G** shown in FIG. **9**, the micro light-emitting diode device **100H** further includes a micro light-emitting diode **120Y**. The micro light-emitting diode **120Y** is also surrounded by the isolation layer **140**, and is configured to emit a yellow light. Since the micro light-emitting diodes **120R, 120G, 120B, 120Y** are surrounded by the same isolation layer **140**, lights emitted by the micro light-emitting diodes **120R, 120G, 120B, 120Y** can be evenly mixed. In some embodiments, a distance of any two of the micro light-emitting diodes **120R, 120G, 120B, 120Y** is less than 2 mm. In this way, lights emitted by the micro light-emitting diodes **120R, 120G, 120B, 120Y** can be mixed more evenly.

(40) According to the foregoing recitations of the embodiments of the disclosure, it can be seen that in the micro light-emitting diode device of the disclosure, the micro light-emitting diode includes the n-doped $\text{Al}_{\text{sub.x}}\text{Ga}_{\text{sub.y}}\text{In}_{\text{sub.(1-x-y)}}\text{N}$ layer that contains aluminum relative to the n-type III-nitride layer, so the refractive index of the n-doped $\text{Al}_{\text{sub.x}}\text{Ga}_{\text{sub.y}}\text{In}_{\text{sub.(1-x-y)}}\text{N}$ layer with the larger band gap will be smaller than the refractive index of the n-type III-nitride layer with the smaller band gap, which can effectively improve the light extraction efficiency of the micro light-emitting diode.

(41) Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

(42) It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A micro light-emitting diode device, comprising: a substrate; a micro light-emitting diode having a lateral width smaller than 100 μm , disposed on the substrate, and comprising: a p-type GaN layer; an n-type III-nitride layer above the p-type GaN layer, wherein III refers to at least one element of aluminum, gallium, and indium; an n-doped $\text{Al}_{\text{sub.x}}\text{Ga}_{\text{sub.y}}\text{In}_{\text{sub.(1-x-y)}}\text{N}$ layer above and in contact with the n-type III-nitride layer, wherein x is equal to or greater than about 0.02, and a thickness of the n-doped $\text{Al}_{\text{sub.x}}\text{Ga}_{\text{sub.y}}\text{In}_{\text{sub.(1-x-y)}}\text{N}$ layer is greater than 10 nm; and an active layer between the p-type GaN layer and the n-type III-nitride layer; and a transparent top electrode at least partially covering and in contact with the n-doped $\text{Al}_{\text{sub.x}}\text{Ga}_{\text{sub.y}}\text{In}_{\text{sub.(1-x-y)}}\text{N}$ layer, wherein a refractive index of the n-doped $\text{Al}_{\text{sub.x}}\text{Ga}_{\text{sub.y}}\text{In}_{\text{sub.(1-x-y)}}\text{N}$ layer is smaller than a refractive index of the n-type III-nitride layer, wherein a sum of thicknesses of the p-type GaN layer, the active layer, the n-type III-nitride layer, and the n-doped $\text{Al}_{\text{sub.x}}\text{Ga}_{\text{sub.y}}\text{In}_{\text{sub.(1-x-y)}}\text{N}$ layer is greater than 400 nm, wherein a sum of the thicknesses of the n-type III-nitride layer and the n-doped $\text{Al}_{\text{sub.x}}\text{Ga}_{\text{sub.y}}\text{In}_{\text{sub.(1-x-y)}}\text{N}$ layer is greater than a sum of the thicknesses of the active layer and the p-type GaN layer.

2. The micro light-emitting diode device of claim 1, wherein the thickness of the n-doped $\text{Al}_{\text{sub.x}}\text{Ga}_{\text{sub.y}}\text{In}_{\text{sub.(1-x-y)}}\text{N}$ layer is greater than 80 nm.

3. The micro light-emitting diode device of claim 1, wherein the thickness of the n-type III-nitride layer is greater than 0.5 μm .

4. The micro light-emitting diode device of claim 1, wherein the thickness of the n-type III-nitride

layer is greater than the sum of the thickness of the active layer and the thickness of the p-type GaN layer.

5. The micro light-emitting diode device of claim 1, further comprising: an isolation layer on the substrate and surrounding the micro light-emitting diode, wherein a height of the isolation layer relative to the substrate is equal to or smaller than a height of a top surface of the n-doped $\text{Al}_{0.5}\text{Ga}_{0.5}\text{In}_{0.5}\text{N}$ layer relative to the substrate.

6. The micro light-emitting diode device of claim 5, wherein a refractive index of the isolation layer is smaller than the refractive index of the transparent top electrode.

7. The micro light-emitting diode device of claim 5, further comprising: a dielectric layer between the isolation layer and a side surface of the micro light-emitting diode.

8. The micro light-emitting diode device of claim 1, wherein the transparent top electrode is made of transparent conductive oxide, and a refractive index of the transparent top electrode is smaller than the refractive index of the n-doped $\text{Al}_{0.5}\text{Ga}_{0.5}\text{In}_{0.5}\text{N}$ layer.

9. The micro light-emitting diode device of claim 1, wherein the transparent top electrode comprises nano metal wires, the transparent top electrode further comprises an encapsulation material in which the nano metal wires are embedded, and a refractive index of the encapsulation material is smaller than the refractive index of the n-doped $\text{Al}_{0.5}\text{Ga}_{0.5}\text{In}_{0.5}\text{N}$ layer.

10. The micro light-emitting diode device of claim 1, further comprising: an isolation layer on the substrate and surrounding the micro light-emitting diode, wherein a height of the isolation layer relative to the substrate is greater than a height of a top surface of the n-doped $\text{Al}_{0.5}\text{Ga}_{0.5}\text{In}_{0.5}\text{N}$ layer relative to the substrate.

11. The micro light-emitting diode device of claim 10, wherein a refractive index of the isolation layer is smaller than the refractive index of the transparent top electrode.

12. The micro light-emitting diode device of claim 10, further comprising: a dielectric layer between the isolation layer and a side surface of the micro light-emitting diode.

13. The micro light-emitting diode device of claim 1, further comprising: an encapsulation layer above and in contact with the transparent top electrode, wherein a refractive index of the encapsulation layer is smaller than the refractive index of the transparent top electrode.

14. The micro light-emitting diode device of claim 1, further comprising: an MgO layer covering and in contact with the transparent top electrode, wherein a refractive index of the MgO layer is smaller than the refractive index of the transparent top electrode.

15. The micro light-emitting diode device of claim 1, further comprising: at least one additional micro light-emitting diode disposed on the substrate, wherein a distance between the micro light-emitting diode and the at least one additional micro light-emitting diode is less than 2 mm; and an isolation layer on the substrate and surrounding the micro light-emitting diode and the at least one additional micro light-emitting diode.

16. The micro light-emitting diode device of claim 15, wherein the micro light-emitting diode and the at least one additional micro light-emitting diode are configured to emit lights of different wavelengths respectively.

17. The micro light-emitting diode device of claim 16, wherein a number of the at least one additional micro light-emitting diode is two, and the lights of different wavelengths comprise a red light, a green light, and a blue light.

18. The micro light-emitting diode device of claim 16, wherein a number of the at least one additional micro light-emitting diode is three, and the lights of different wavelengths comprise a red light, a green light, a blue light, and a yellow light.
